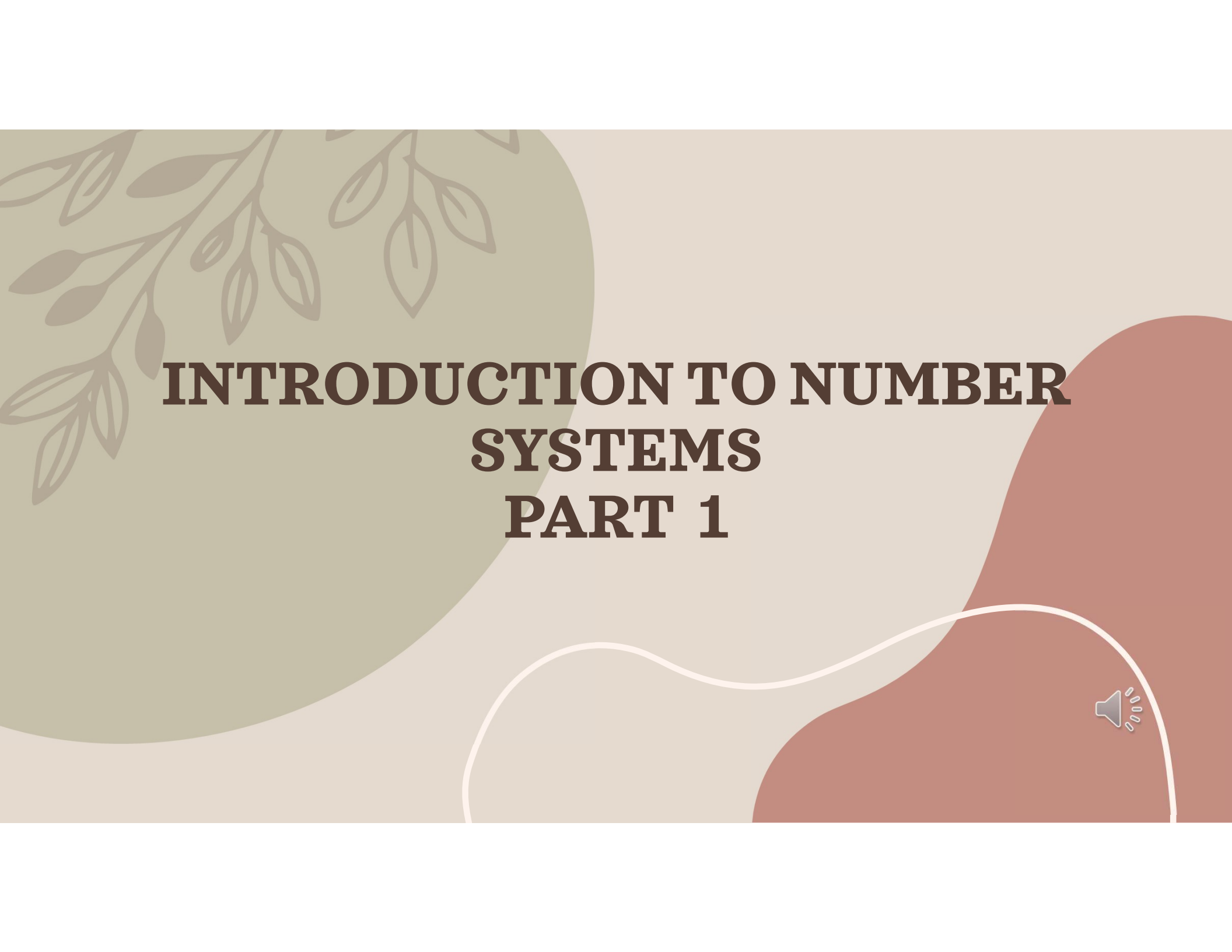




INTRODUCTION TO NUMBER SYSTEMS PART 1



TOPICS

1. TYPES OF NUMBER SYSTEMS

2. WEIGHTED AND
UNWEIGHTED NUMBER
SYSTEMS



NUMBER SYSTEM

Number system is defined as a set of values used to represent a quantity

Number System	Radix (Base)
Binary	2
Octal	8
Decimal	10
Duodecimal	12
Hexadecimal	16

0,1	101101 ₂
0,1,2,3,4,5,6,7	735 ₈
0,1,2,3,4,5,6,7,8,9	458 ₁₀
0,1,2,3,4,5,6,7,8,9,A,B	2A7 ₁₂
0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F	3F9 ₁₆

Number of bits = 0 to (r-1)

Example 1

For binary number system , radix =2

There fore , number of bits = 0 to (2-1) = (0 to 1)

Example 2

For decimal number system , radix = 10

There fore , number of bits = 0 to (10-1) (0 to 9)



WEIGHTED AND UN WEIGHTED NUMBER SYSTEM

In a weighted number system, each digit position has a weight based on the radix (base).

For example , 542_{10}

2 = units place

4 = tens place

5 = hundreds place

Digit	Weight	Value
5	$10^2 = 100$	500
4	$10^1 = 10$	40
2	$10^0 = 1$	2

- Binary → powers of 2
- Octal → powers of 8
- Decimal → powers of 10
- Hexadecimal → powers of 16

A number system in which digit positions do not have fixed weights and symbols represent values directly.

For example , take a roman number system

X = 10

V = 5

I = 1

XVI = 16

Note

As radix increases the number of bits represents the system decreases

For example

$$(7392)_{10} = (1110011100000)_2$$

$$(7392)_{10} = (16340)_8$$

$$r_1 < r_2, N_1 \geq N_2$$



Examples for weighted and unweighted number systems

Weighted number systems

1. Binary number system
2. Octal number system
3. Hexa decimal number system
4. Decimal number system

Unweighted number systems

1. Grey codes
2. Excess three codes





thank you